

csiro-biomass

Synthetic dataset

Rank	2
Team	dino series
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Source: <https://www.kaggle.com/competitions/csiro-biomass/discussion/670664>

Retrieved with Kaggle CLI on 2026-07-18.

This is the synthetic dataset I generated from the training set using [Qwen-Image-Edit](#):

<https://www.kaggle.com/datasets/quan0095/csiro-synthetic-dataset>

One of the main challenges in using synthetic datasets for image regression is the ground truth issue; when an image is modified, the biomass value of each target changes accordingly. To address this, I employed a pseudo-labeling strategy: I used a model trained on the original training set to generate labels for the synthetic dataset, and subsequently trained the final model on the combined dataset (original training set + synthetic dataset).

- The synthetic dataset was generated by simulating seasonal weather changes and adding artifacts such as gravel and rocks.
- I removed all grass vegetation from the images, retaining only the bare soil, and set the target values for clover, dead, and green to 0. These images were then combined with the training dataset to help the model discriminate between biomass-containing regions and non-biomass regions.

Supplemental Q&A

Kh0a · 2026-01-29T04:01:43.363000

very interesting, i did try pseudo-labeling on these datasets: <https://www.kaggle.com/datasets/usharengaraju/grassclover-dataset>, <https://www.kaggle.com/datasets/amirmmahdavia/irish-grass-clover/data>. But not very successful

Quan Vu · 2026-01-29T04:04:58.380000

I attempted to pretrain on this dataset, but the results were unsatisfactory. There is a significant domain gap between the two datasets: they differ in crop size and grass types, and the labeling protocols may also vary.

Mohamed Eltayeb · 2026-01-29T06:27:26.397000

I used Banana Pro to convert all green in the images into dead, then to ensure correct targets i just set green = clover = gdm = 0, and dead = total = dead+green+clover but it didn't work :)

Quan Vu · 2026-01-29T08:17:31.423000

We need to limit the ratio of zero labels to avoid distorting the original distribution. I only introduced roughly 10 images with a label of 0 (totaling ~10/357).

Quan Vu · 2026-01-29T08:22:46.880000

Applying synthetic data to biomass estimation presents more challenges than classification. In classification, the class is invariant to augmentation (e.g., a cat remains a cat), whereas in image regression, visual modifications inherently alter the target values. I initially attempted to train using synthetic data with the original labels, but this led to a decrease in LB performance.

Komil Parmar · 2026-01-29T09:06:46.907000

Aahhh, Pseudo labelling. Just as I had predicted. Great approach. I actually thought of using Nano banana for this, but then thought like how many samples will I make one by one online. May I know how did you use Qwen? Like installing it on local machine or on colab, and was there any pipeline involved?

Quan Vu · 2026-01-29T09:26:35.990000

I'm using a local setup, and in the dataset I've attached the scripts that I used to generate the data

Quan Vu · 2026-01-29T09:32:01.330000

I tried using Nano banana to generate images with very good quality, but Qwen is a better choice when you need a local setup and to generate a large number of images.